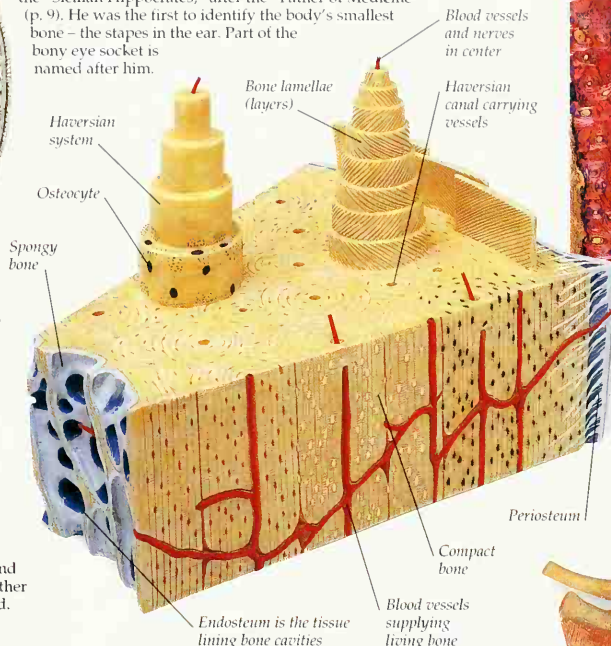




BONE EXPERT

Giovanni Ingrassias (1510-1580) was an early osteologist, or bone expert. He was physician and anatomy professor at Naples and then Palermo, where he gained a reputation as the "Sicilian Hippocrates," after the "Father of Medicine" (p. 9). He was the first to identify the body's smallest bone – the stapes in the ear. Part of the bony eye socket is named after him.



RINGS AND LAKES

Under the light microscope, a very thin slice of bone has been dyed with a chemical stain to show its features.

On the left are the circular Haversian systems of compact bone, looking like growth rings in a tree trunk.

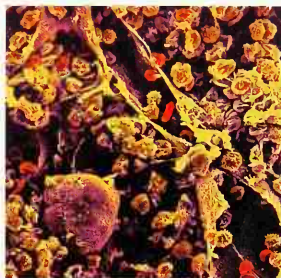
In the centers are tiny lacunae or lakes of blood. To the right is the more open structure of spongy bone.

BONE MICROSTRUCTURE

Each tube in the Haversian systems in compact bone is about 0.01 in (0.25 mm) wide, and made of layers of bone, called lamellae. In its center are tiny blood vessels and nerves. Embedded in the lamellae are osteocytes – the cells that make and maintain the living bone, and which survive for 20 years or more. Bone itself consists of two main components. Fibers of the body protein collagen, also found in the skin and connective tissues, make the whole structure slightly flexible and resilient. In addition, crystals and salts of calcium, phosphorus, and other chemicals make bone hard and rigid.

Bone marrow

In all bones of a baby, and in some adult bones, jellylike marrow occupies the central cavity. Its main job is to make new blood cells, as these die rapidly and need replacing. Every second, more than 2 million red cells leave the marrow to enter the blood's circulation, in the vessels weaving through the bone structure. The marrow and bone also act as a mineral reserve. If certain nutrients are in short supply, these are taken out of the bone and sent to other, more vital sites.



GROWING BLOOD CELLS

Making blood cells – hematopoiesis – is a complex process. In early life, "parent" stem cells in the red marrow multiply to produce "committed" stem cells. These divide and their "offspring" mature rapidly to make red blood cells by the billion, and four main types of white blood cells (p. 33).

RIBS AND BREASTBONE

The ribs and breastbone, or sternum, are especially well equipped with bone marrow. Blood-cell-making red marrow is present in all of a baby's bones, but only remains in the adult's skull, spine, breastbone, ribs, shoulder, and hip bones. During teenage years, it is gradually replaced in the limbs by yellow marrow, which has more fatty tissue.

